WMO Data Rescue Initiative
Why it matters now?

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ESMO SSG Annual Meeting and WGORC Kick Off Meeting
December 2025

Why Data Rescue is a Top Global Priority?

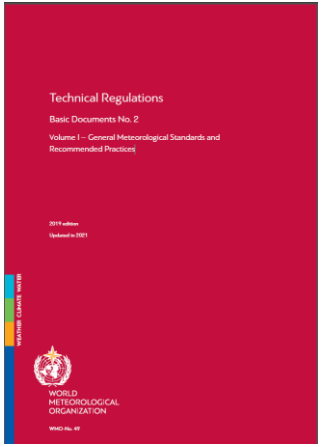
- Mandatory under the updated WMO Regulations (planned for 2026)
- Safeguards **quality**, **continuity** and **completeness** of climate datasets
- Essential for climate **monitoring**, **services**, and **historical reconstruction**

Data Rescue is now recognized as a foundational element of climate data stewardship. It ensures that critical observational records – some of them irreplaceable – remain accessible, usable and internationally exchangeable

WMO Regulatory Evolution (2026)

- Data Rescue added to WMO Technical Regulations **WMO-No. 49**

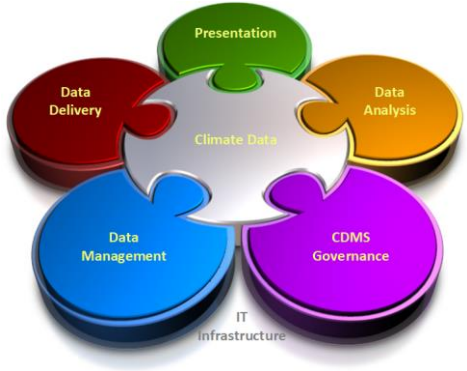
- Manual **WMO-No. 1238** requires Members to:
  - ✓ Develop a national Data Rescue **strategy**
  - ✓ Preserve all climate data
  - ✓ Maintain duplicate repositories
  - ✓ Coordinate & communicate Data Rescue activities



These obligations elevate Data Rescue to a **formal component of the WMO Global Data Management Framework**, strengthening accountability across all National Meteorological and Hydrological Services (**NMHSs**)

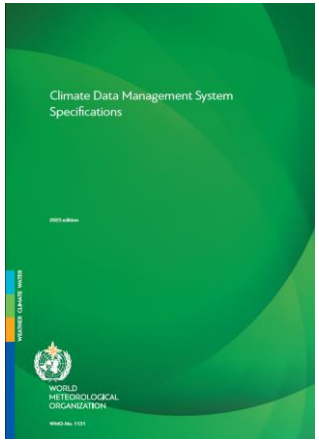
Climate Data Management System (CDMS) Compliance

WMO Members should verify that their CDMS implementation satisfies the **Required** Data Rescue specifications.
These specifications define what is necessary for full WMO compliance across Data Governance, Timeseries, Management, Analysis, Delivery, and Presentation



High Level Components of a CDMS

- 2025, CDMS Specifications **WMO-No. 1131**
- Components classified as **Required** / **Recommended** / **Optional**
- Members are encouraged to fully implement **required** **Data Rescue components**



WMO-C3S Data Rescue Portal

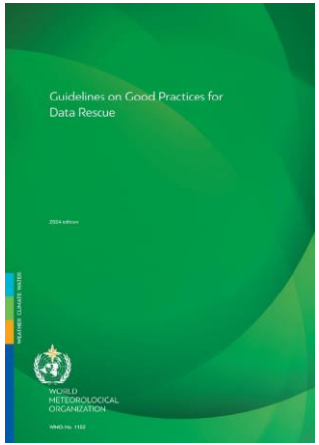
150+ projects already visible

- Share projects
- Avoid duplication
- Request assistance
- Identify potential funding partners

The Portal has become the central coordination platform for the global Data Rescue community

WMO Data Rescue Guidelines on Good Practices

- 2024 Edition of **WMO-No. 1182**, Unified multi-domain guidance
- **Integrates** WMO **Hydrology, Marine and Land Data**
- **Incorporate Copernicus** (C3S) **Guidelines**
- Provides standardised workflows, quality expectations, and metadata requirements



This publication is now the **reference standard** for operational Data Rescue across NMHSs

Capacity Building / Angola Project

- Over the past decade, WMO has provided support for Data Rescue to more than **28 NMHSs**, including those in Africa, Asia, the Caribbean, and South America

- **2024: Angola Data Rescue Pilot**
  - ✓ Development of an operational national plan
  - ✓ Multi-partner collaboration: **INAMET**, **SADC**, **WMO**, **KNMI**, **IEDRO**, **IDEMS**, **DWD** and **MFI**

➡ All climate-related projects are encouraged to include a **dedicated Data Rescue component**

International Case Studies

- **UK Rainfall Rescue** – 16.000 volunteers, 5.4M digitized observations
- **Italian Precipitation Records** – Student-driven citizen science (1921-1950)
- **Irish Weather Rescue** – 39% completed (1860–1950)
- **Copernicus African Data Initiative** – 4 million scanned pages across 41 countries & **Data Rescue in the Classroom** – Expanding student involvement

Citizen science and students engagement have become powerful accelerator for large-scale Data Rescue activities

Data Rescue Community Actions Plan

- Extend long climate series **back to 1850**
- Complete the period **1991-present**
- Include **Marine** and **Hydrological observations**
- Strengthen WMO **Regional Climate Centres** (RCCs) roles
- Encourage NMHSs to populate their metadata in WMO **OSCAR/Surface** extensively
- Promote the establishment of WMO **Global Historical Climate Data Centres**

These actions reflect the priorities collectively identified by the global Data Rescue community and endorsed at recent WMO workshops (e.g. Astana 2025)

Conclusion

Data Rescue is a continuous, long-term investment that underpins climate monitoring, climate services and historical reconstruction

The standards and tools now exist

What we need is **wider engagement**, stronger coordination, and systematic integration into national, regional and global climate programmes

Let us mobilise our communities, digitise our archives, and ensure that these irreplaceable climate records are preserved, shared and used openly

Merci!

In the following slides:

- Links to upload your climate data to open portals
- Useful links for Workshop, Forum, etc.
- Graphic NMHS and WMO systems
- Data Rescue requierements from WMO-N.1131
- Acronyms used
- Data Rescue Definition
- How WMO Members can contribute
- International Case Studies links

Upload your climate data to open portals

Share your data via Copernicus
https://datadeposit.climate.copernicus.eu/home/

Share your data via ECA&D (Europe RCC for data)
https://www.ecad.eu
eca@knmi.nl

Share your data via NOAA
https://www.ncei.noaa.gov/archive/send2ncei/

Share your data via PANGAEA, Data Publisher for Earth & Environmental Science
https://www.pangaea.de/

Share your data via Zenodo, Open repository for research outputs
https://zenodo.org/

Useful links for Workshop, Forum, etc.

- Eumetnet, Data Management Workshop (Oslo, 11/2025)
- ACRE, Atmospheric Circulation Reconstructions over the Earth (Rome, 09/2026)
- EGU, European Geosciences Union, Historical Weather Data Rescue and Methodologies Focusing on Data-sparse Regions (Vienna, 5/2026)
- IEDRO, International Environmental Data Rescue Organization

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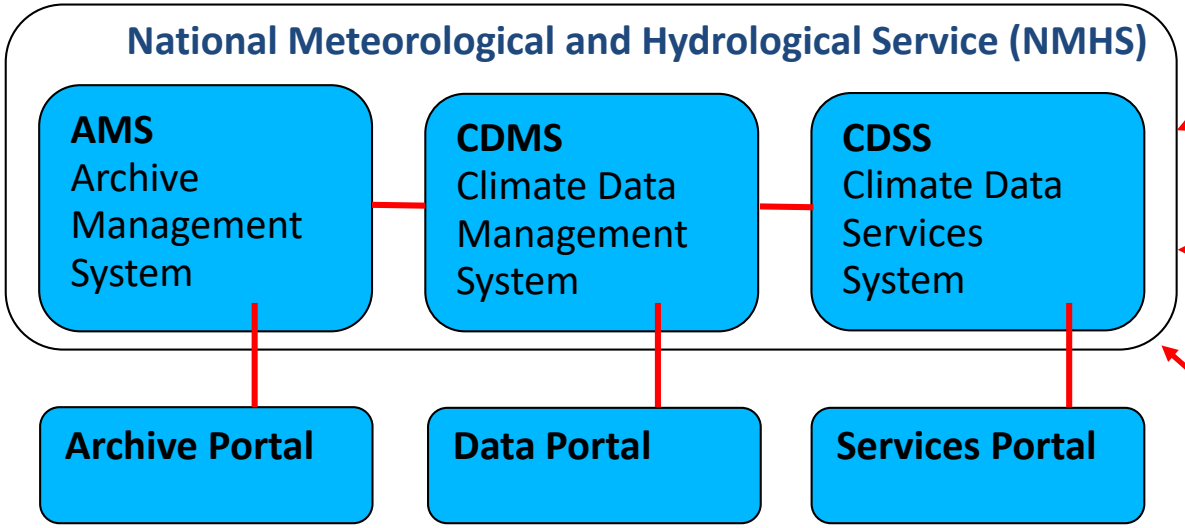
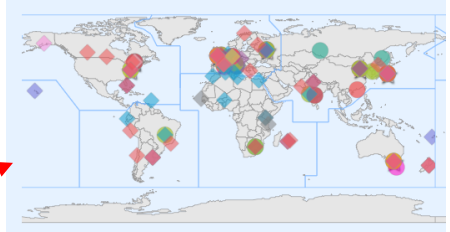
Liberté
Égalité
Fraternité

METEO FRANCE
À VOS CÔTÉS, DANS UN CLIMAT QUI CHANGE

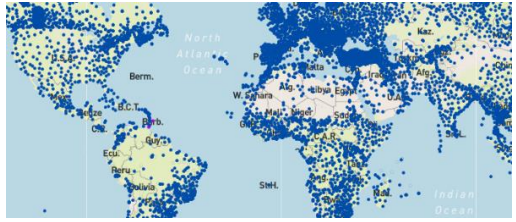
NMHS and WMO tools



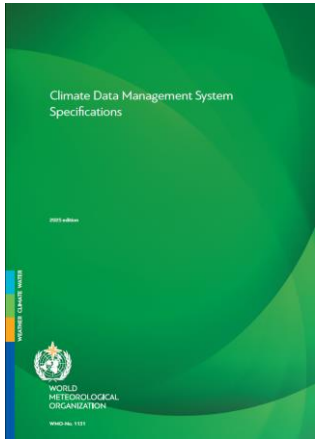


WMO WIPPS

WMO Data Rescue Portal

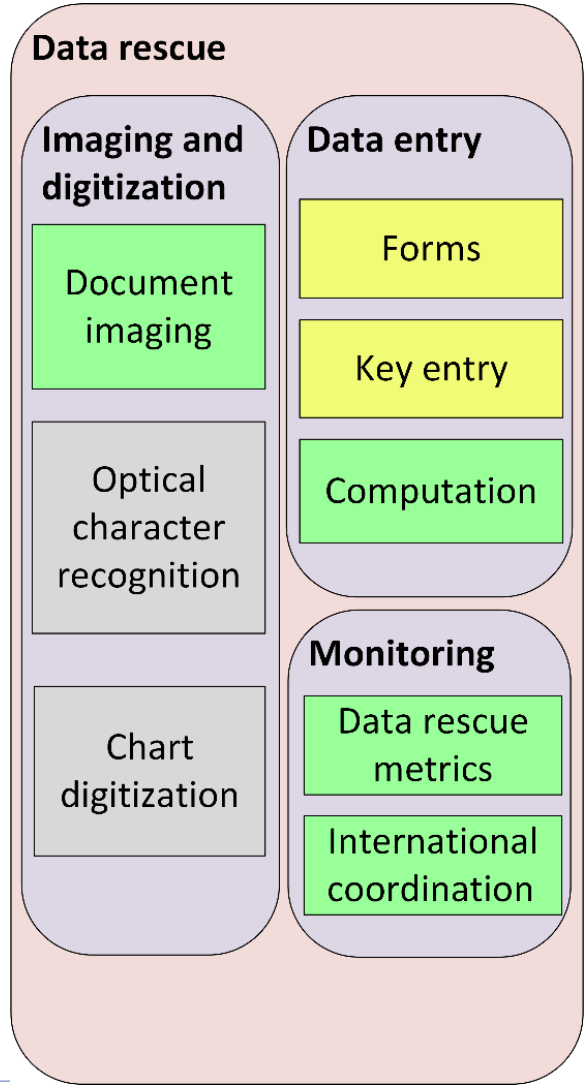


WMO OSCAR Metadata



Climate Data Management System (CDMS) Specifications

- 2025, **CDMS Specifications WMO-No. 1131**
- Components classified as **Required** / **Recommended** / **Optional**
- Members are encouraged to fully implement the **required Data Rescue components**
- **ET-DDS Wiki:** https://github.com/ET-DRC/WMO-No.1131/wiki



Acronyms

C3S	Copernicus Climate Change Service
CDMS	Climate Data Management System
DWD	German Meteorological Service
ET-DDS	WMO Expert Team on Data Development and Stewardship
IDEMS	Impact commercialization partner to open-source mathematics assessment platform
IEDRO	International Environmental Data Rescue Organization
INAMET	National Meteorological Service of Angola
KNMI	Royal Netherlands Meteorological Institute
MFI	Meteo France International
NMHS	National Meteorological an Hydrological Service
OSCAR/Surface	WMO Observing Systems Capability Analysis and Review Tool
RCC	Regional Climate Center
SADC	Southern African Development Community

Data Rescue Definition

Data rescue – WMO-No. 1238

The ongoing process of identifying and preserving all data, records and climate archives that are at risk of being lost, and of digitizing current and past data into computer-compatible form for easy access.

The identification process also needs to search for data possibly contained in non-NMHS repositories such as universities, libraries and national archives. In some cases historical data may be held in a country different to that of the Member concerned.

Data rescue also includes migration from obsolete or computer-corrupted media to modern media and readable formats. This definition applies to all meteorological and hydrological data

How Members Can Contribute

- Develop/update Data Rescue strategy
- **Inventory**, document and **share information** on historical archives
- **Deposit digitised datasets in recognised open repositories**: Copernicus, NOAA, ECA&D, PANGEA, ZENODO, …
- Engage citizen/classroom
- Use WMO Data Rescue Portal
- **Ensure full and accurate metadata submission in WMO OSCAR/Surface**

International Case Studies

- **UK citizens extend rainfall records**, https://www.zooniverse.org/projects/edh/rainfall-rescue
- **Italian precipitation records**, https://journals.ametsoc.org/view/journals/bams/106/3/BAMS-D-24-0078.1.xml
- **Irish Weather Rescue**, https://www.zooniverse.org/projects/met-rhonda/irish-weather-rescue
- **African data**, https://www.zooniverse.org/projects/healion90/weather-archive-africa, also with student participation ***Data Rescue in the Classroom*** (https://doi.org/10.1002/gdj3.248)